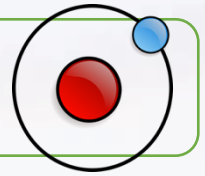
Lecture 16
Simulation Based Inference for Astrophysics and Cosmology

Lecturer: **Dr Harry Bevins** (htjb2)

An SKA observation

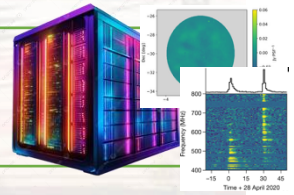
L1:
Physics



L 2,4,5:
Instrument

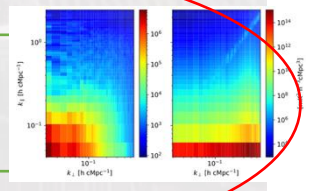
L6:
Calibration

L7-10:
Science data



L11-12:
Data handling

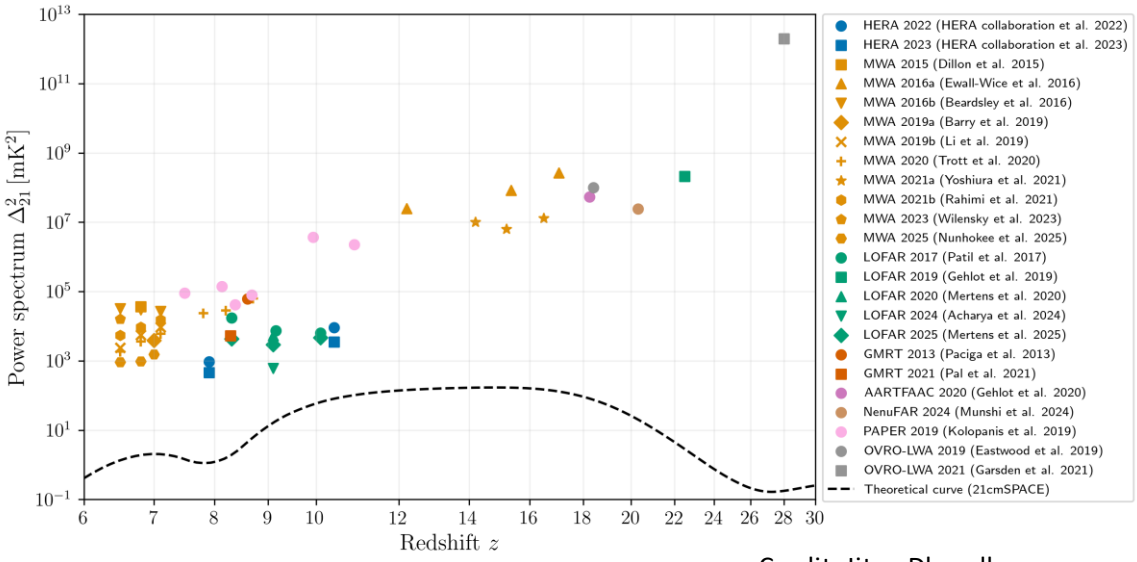
L13-16:
Science



Lectures 13-16: Science extraction

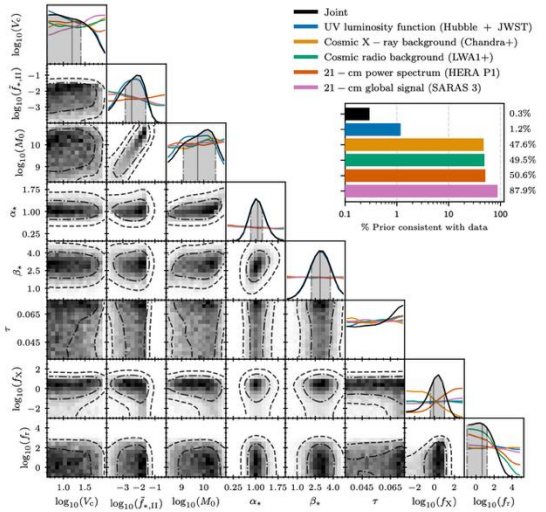
Example:

The Epoch of Re-ionization and the 21-cm line



Credit: Jiten Dhandha

From **science data products**



Dhandha et al. 2025

To **science**

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Where we left off last lecture…

- Why not just emulate the whole likelihood?

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Overview

- What is Simulation Based Inference?
- Approximate Bayesian Computation
- Neural Posterior Estimation
- Neural Ratio Estimation

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What is Simulation Based Inference?

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Why not emulate the likelihood?

$$\log_e L(\theta) = \sum_i -\frac{1}{2}\log_e|2\pi\Sigma| - \frac{1}{2}\left(D_i - M_i(\theta)\right)^T \Sigma^{-1}(D_i - M_i(\theta))$$

- In the last lecture I discussed how the likelihood defines the noise distribution we expect in our data
- But often this is hard to know and approximations like the above Gaussian don't really describe our data well
- We looked at emulating the model describing our data but we can also think about emulating our likelihood function

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Simulation Based Inference (SBI)

- Sometimes referred to as Likelihood-free Inference (not a good name)
- Or Implicit likelihood inference (a better name)
- The idea is to learn an approximation for the likelihood or indeed the posterior via simulations of the data
- Simulations here includes the physics we are interested in, and contaminating signals, the noise and the impact of the instrument we are observing with

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Simulation Based Inference (SBI)

- A number of different Simulation-Based inference algorithms
- The idea has been around for a long time but has become more feasible with the advent of advance machine learning tools
- Examples include; Approximate Bayesian Computation, Neural Posterior, Neural Likelihood, Neural Joint and Neural Ratio estimation
- See K. Cranmer et a. [1911.01429] for a review

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Simulation Based Inference (SBI)

- Beneficial when
  - we can not analytically write the likelihood
  - we do not know the noise distribution in our data
  - the analytic likelihood is too computationally expensive

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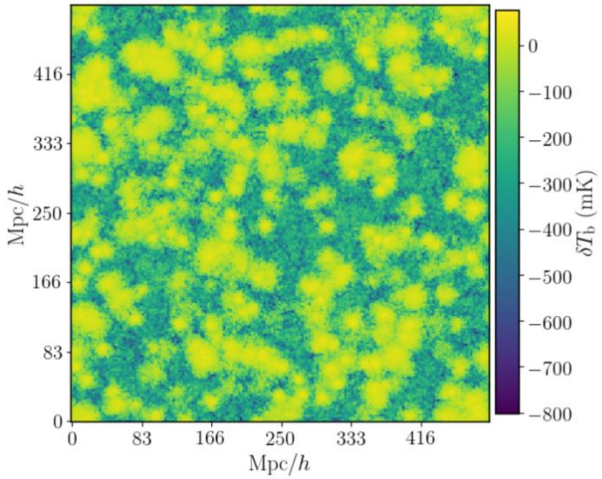
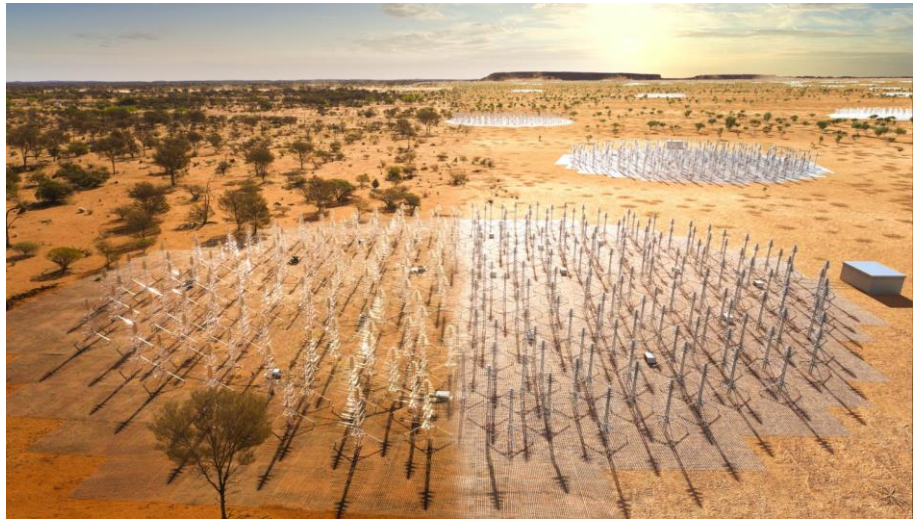
Field Level Inference

- What cosmological parameters give us this 21-cm intensity map?
- Could compress to summary statistics but there is potential loss of information
- Hard to write an analytic likelihood
- Turn to SBI instead



Bera et al. 2022 [2210.12164]
$z \approx 9$



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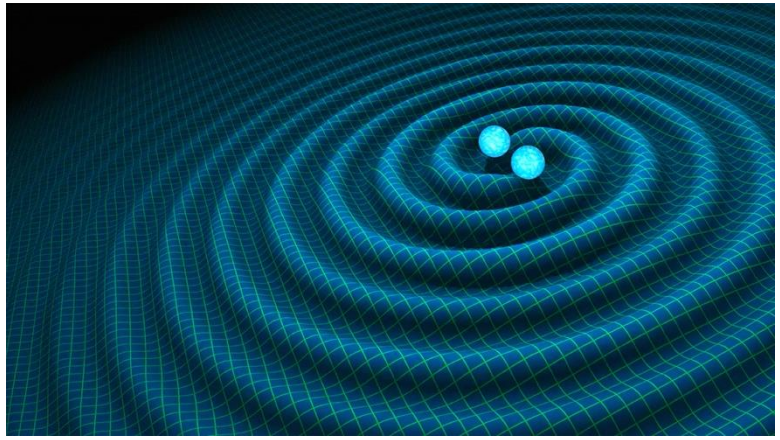
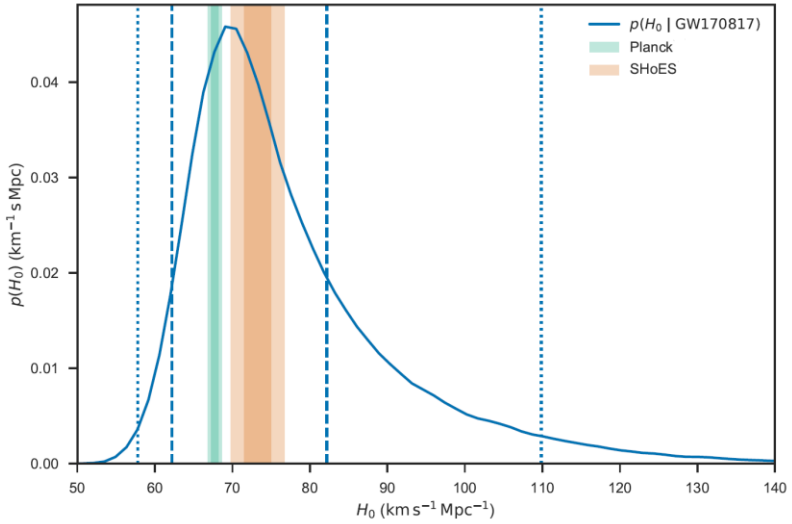
GW Follow Ups

- GW chirps give us distance
- Electromagnetic follow up observations of GWs can help us pinpoint the redshift of the sources
- With distance and redshift we can learn about the expansion rate of the universe
- But GWs emitted from cataclysmic events with short lifetimes so follow ups have to be triggered quickly
- SBI allows us to do that





[1710.05835]

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Approximate Bayesian Computation

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The idea

- ABC originates from the 80s (Rubin 1984) and was one of the first proposed techniques for SBI
- The idea is to generate a set of simulations covering a broad prior parameter space
- Define a distance metric between the data and simulations $\rho(D, S)$
- And define a non-zero tolerance ϵ such that samples are accepted into the posterior based on $\rho(D, S) < \epsilon$

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Approximating the posterior

- Effectively we are approximating $P(\theta|D)$ with $P(\theta|\rho(D,S) < \epsilon)$
- For small ϵ this approximation is usually okay
- However, if ϵ is set too large then the approximation will be poor
- What constitutes small and large will be data specific and so setting ϵ is hard

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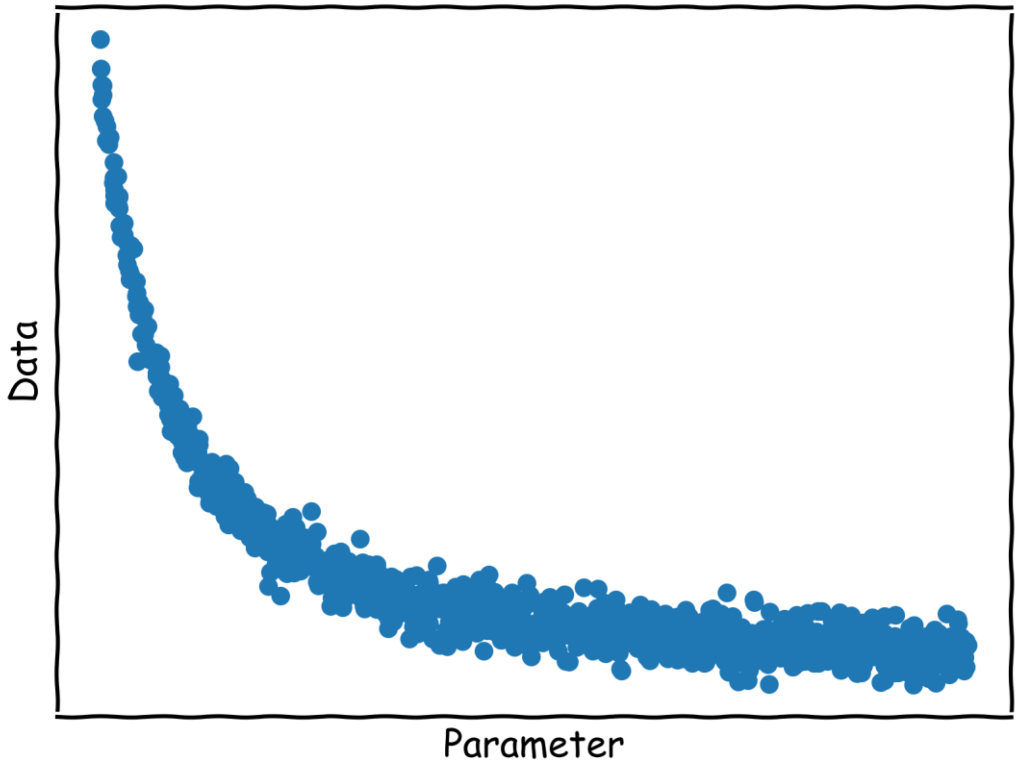
An Example

- Single parameter problem such that our data

$$D = \theta^{-1.5} + \sigma$$

- Where σ is Gaussian random noise

- Think about the joint distribution $P(D, \theta)$



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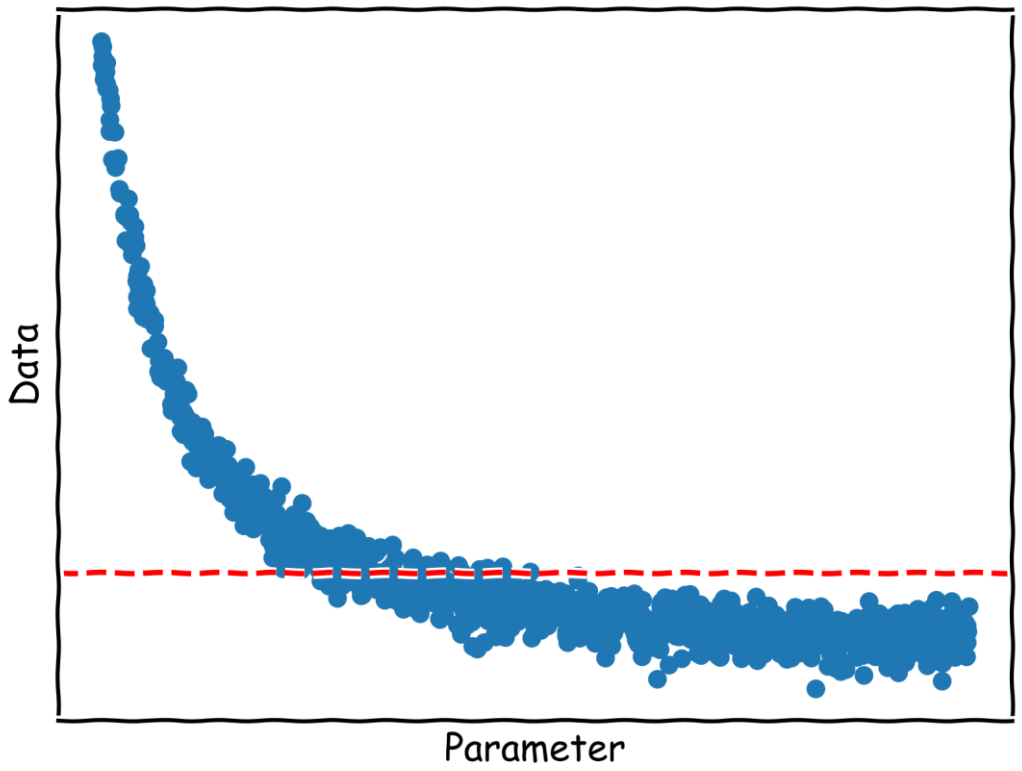
An Example

- Make an observation of our data
- Define a distance metric

$$\rho(D, S) = |D - S|$$

- Pick a value for epsilon say 20



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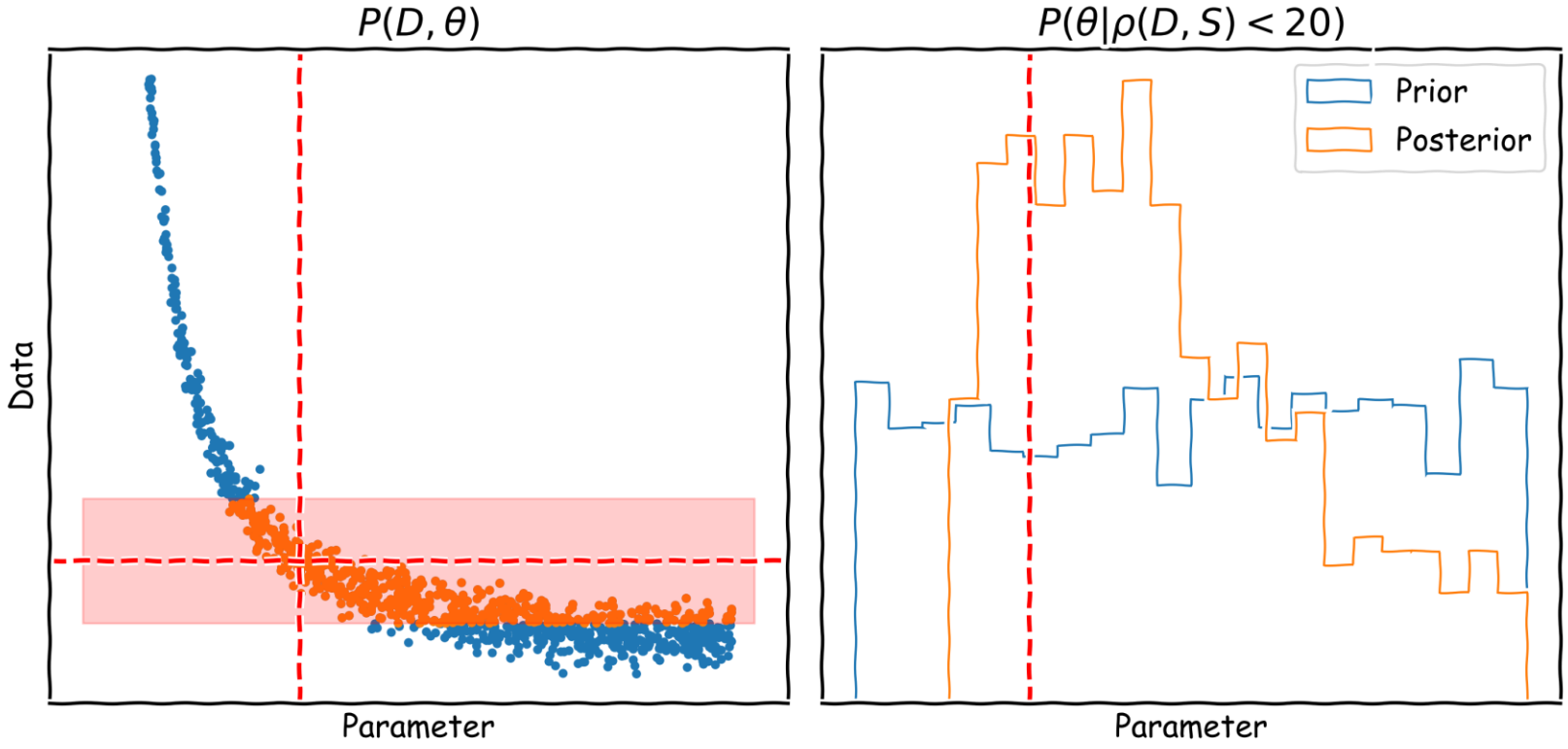
$\epsilon = 20$



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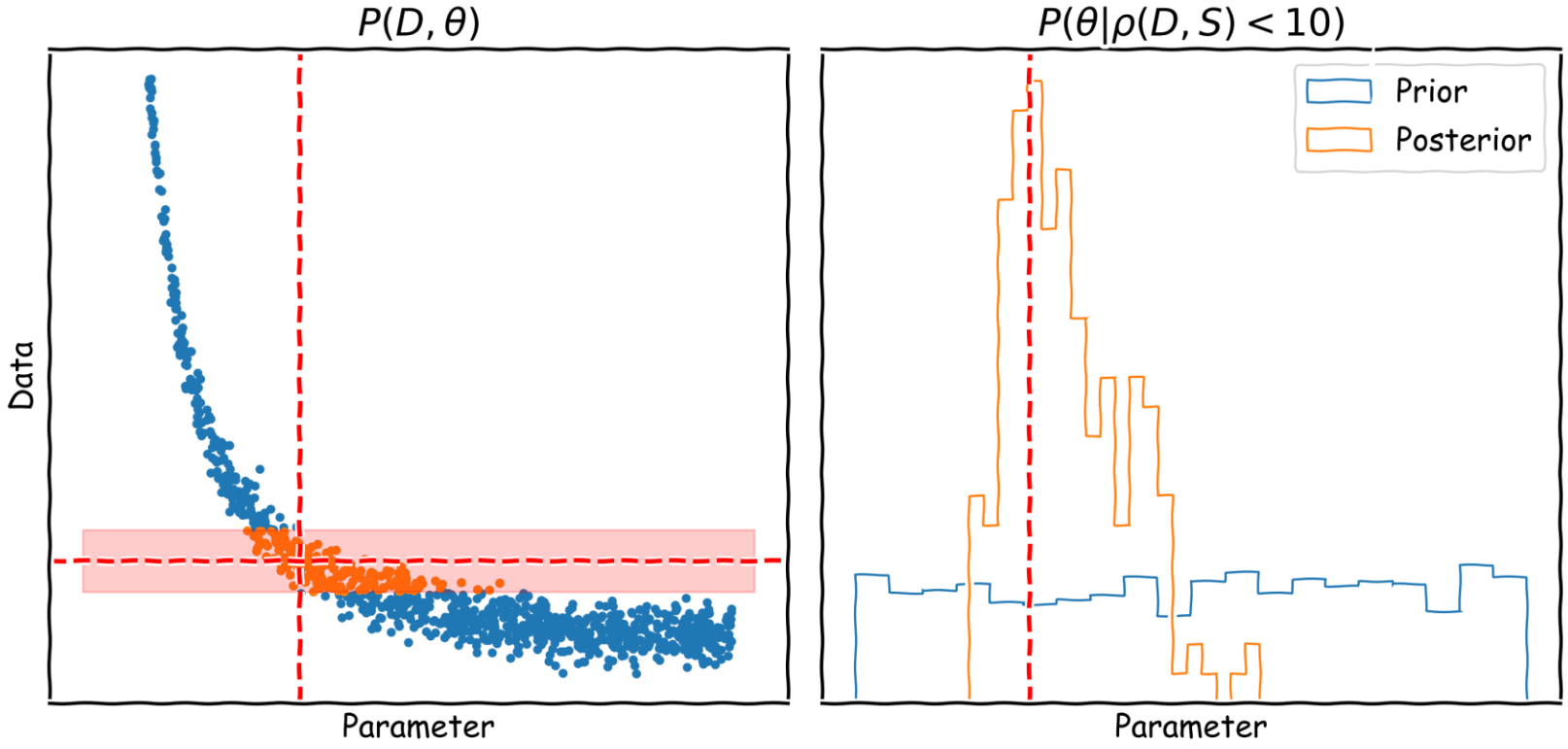
$\epsilon = 10$



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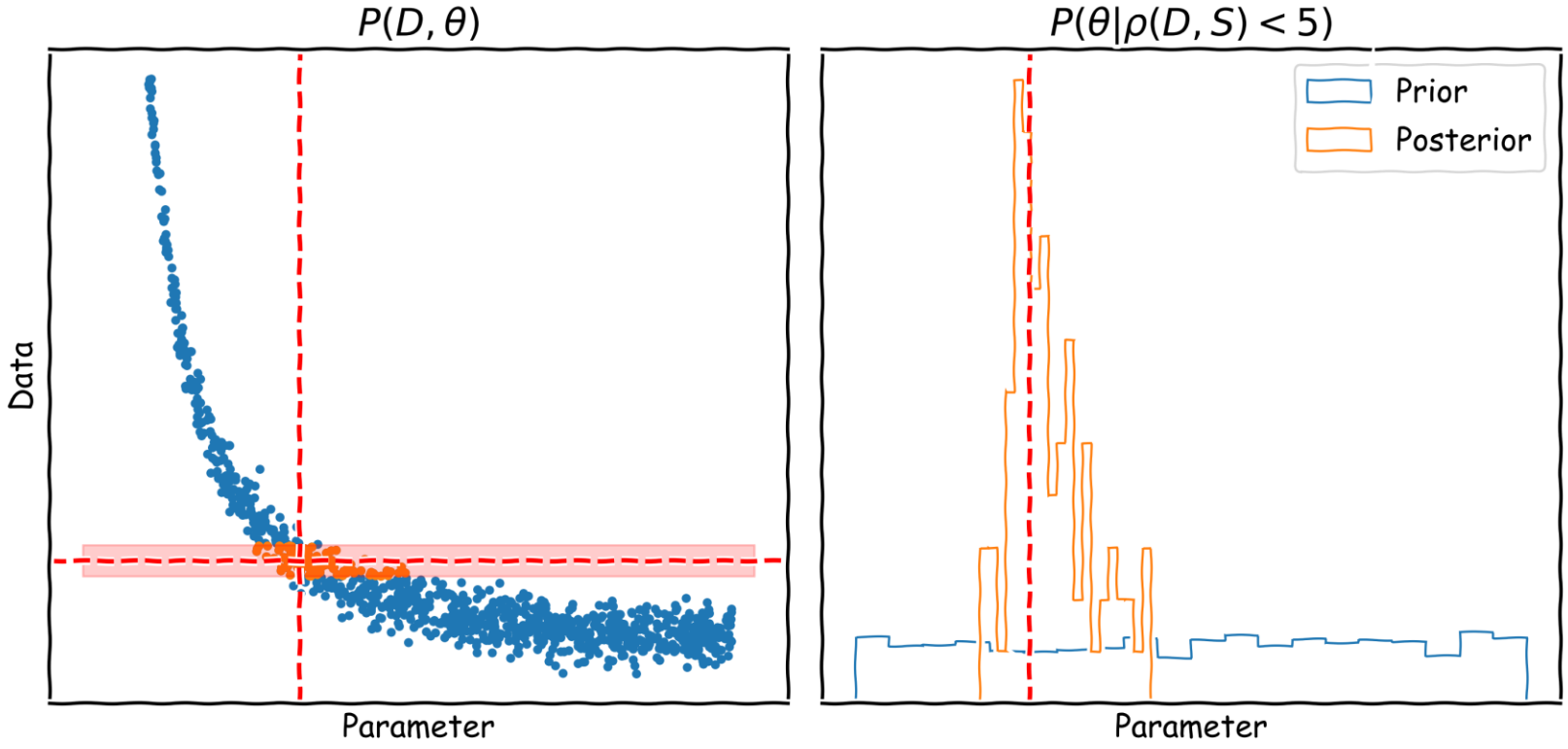
$\epsilon = 5$



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What does the algorithm look like?

Distance Metric

Prior samples and simulations

```
def distance(sim_d, d):
    return np.abs(d - sim_d)

def sampling(samples, data, epsilon):

    posterior = []
    accepted_data = []
    for i in range(len(samples)):
        if distance(data[i], true_data) < epsilon:
            posterior.append(samples[i])
            accepted_data.append(data[i])
    posterior = np.array(posterior)
    accepted_data = np.array(accepted_data)
    return posterior, accepted_data
```

$\rho(D, S) < \epsilon$

Accepted samples and simulations

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No likelihood calls

- We did not make a single call to any likelihood function here
- But there is a cost to generating our simulations
- We could use emulators to generate the simulations if they are prohibitively costly
- Typically need fewer simulation calls than during an MCMC run
- Likelihood might be intractable anyway so MCMC might not be an option

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Problems with ABC?

- How do we set ϵ?
- Becomes very expensive when we have many dimensions and big data sets (curse of dimensionality)
- Sensitive to accuracy of simulations (more general for SBI)
- Not amortized? Basically, means that if our data changes we have to repeat most of the process

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Neural Posterior Estimation (NPE)

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What is NPE?

- The idea is to build some model for $P_\phi(\theta|D)$ with parameters ϕ
- Then show the model examples of data and parameters to optimize ϕ
- The posteriors are often modelled with neural networks known as Normalising Flows (NFs), but you could use GPs, diffusion models…
- After an initial cost to generate simulations and train the NF we can apply the algorithm to many different data sets

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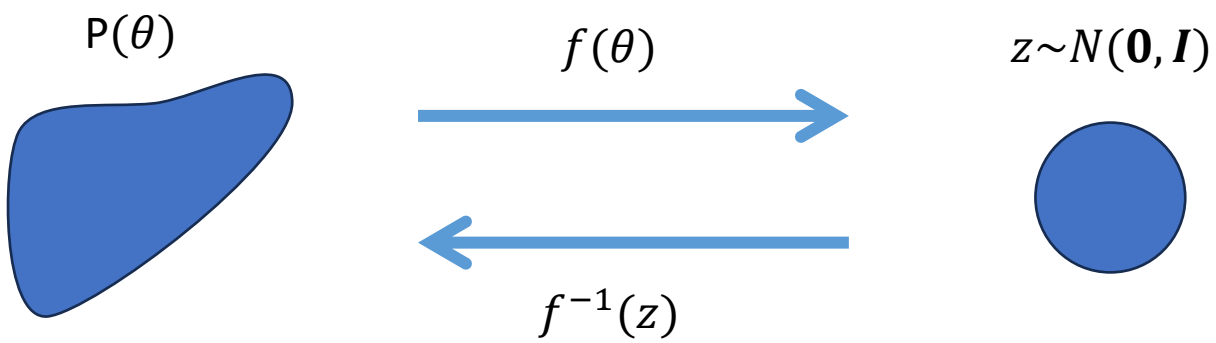
Density Estimation with Normalizing Flows

- Invertible transformation from some known distribution to a more complex target distribution

- Many different flavours like NICE, RealNVP and MAFs



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Masked Autoregressive Flows

- We choose to represent our data as a series of conditionals

$$P(\theta) = \prod_{i=1}^{N} P(\theta_i | \theta_1, \theta_2, \dots, \theta_{i-1})$$

- And model each conditional as a Gaussian distribution

$$P(\theta_i | \theta_1, \theta_2, \dots, \theta_{i-1}) = N(\mu_i, \sigma_i)$$

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Masked Autoregressive Flows

$$P(\theta_i|\theta_1, \theta_2, \dots, \theta_{i-1}) = N(\mu_i, \sigma_i)$$

- Our conditionals are just a shifted and scaled version of the base distribution (standard normal)

$$\theta_i = \mu_i + \sigma_i \,.\, z_i, \;\; z_i \sim N(0, 1)$$

- μ_i and σ_i are functions of the neural network hyperparameters and the actual output of the neural network

$$f_\phi(\theta) \;\rightarrow \mu(\phi), \sigma(\phi)$$

and so we optimize them using some loss function

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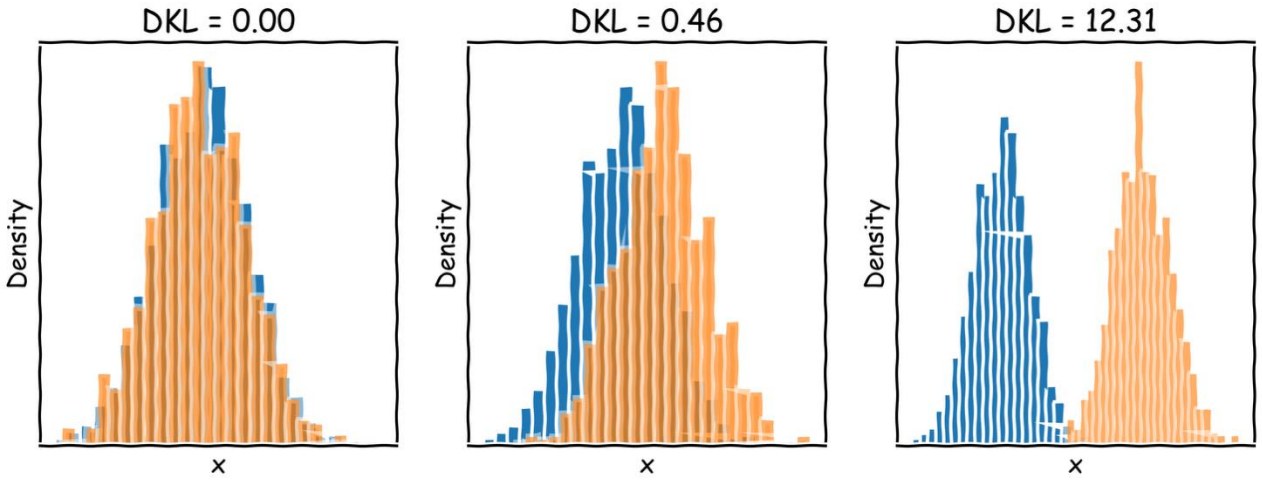
Loss Function

- Minimize the KL divergence between the probability of some samples on the network $P_\phi(\theta)$ and the true target probability $P(\theta)$

$$D_{KL} = \int P(\theta) \log \frac{P(\theta)}{P_\phi(\theta)} d\theta$$



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Loss Function

$$D_{KL} = \int P(\theta) \log \frac{P(\theta)}{P_\phi(\theta)} d\theta$$

$$D_{KL} = \int P(\theta) \log P(\theta)\, d\theta \; - \int P(\theta) \log P_\phi(\theta)\, d\theta$$

- The first term is independent of ϕ so we can ignore it, and the last term is just

$$\int P(\theta) \log P_\phi(\theta)\, d\theta = < \log P_\phi(\theta) >_{P(\theta)} = \frac{1}{N} \sum_j \log P_\phi(\theta_j)$$

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Loss Function

- To evaluate $\log P_\phi(\theta_j)$ we use the change of variables formula between the target and the standard normal distribution

$$\log P_\phi(\theta) = \log N\left(f_\phi(\theta)|\mathbf{0}, \boldsymbol{I}\right) + \log(\det\left|\frac{d f_\phi(\theta)}{d\,\theta}\right|)$$

where $f_\phi(\theta)$ is the neural network

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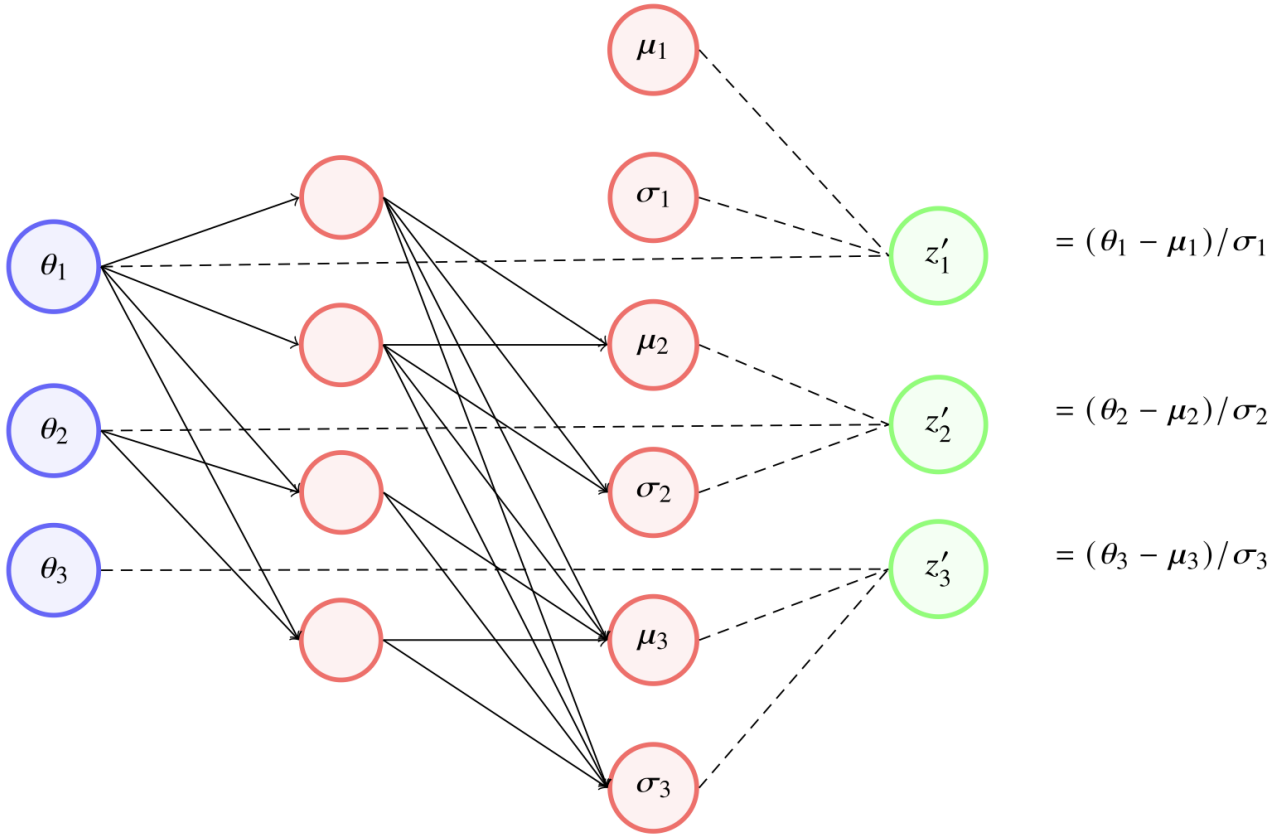
The Neural Network



Bevins et al 2023

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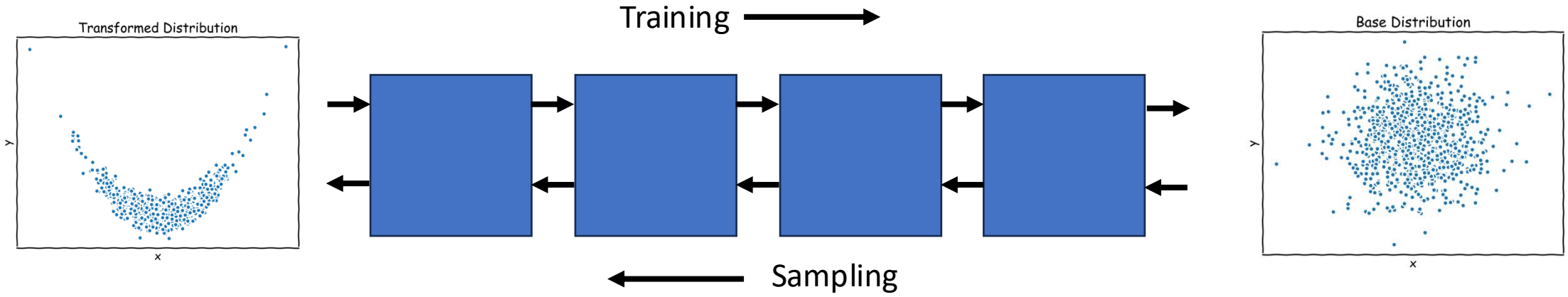
Masked Autoregressive Flow



- Chain a series of these NNs together to get MAFs
- Idea is that we can learn more complex distributions with more complex architectures
- Essentially shifting and scaling repeatedly the base distribution until we get something like the target
- The modelled conditionals in each NN will be slightly different

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But this is just density estimation?

- NFs estimate the probability of $P(\theta)$?
- We really want is a posterior distribution i.e. $P(\theta|D)$
- To do this we use conditional MAFs

$$P(\theta|D) = \prod_{i=1}^{N} P(\theta_i|\theta_1, \theta_2, \dots, \theta_{i-1}, D)$$

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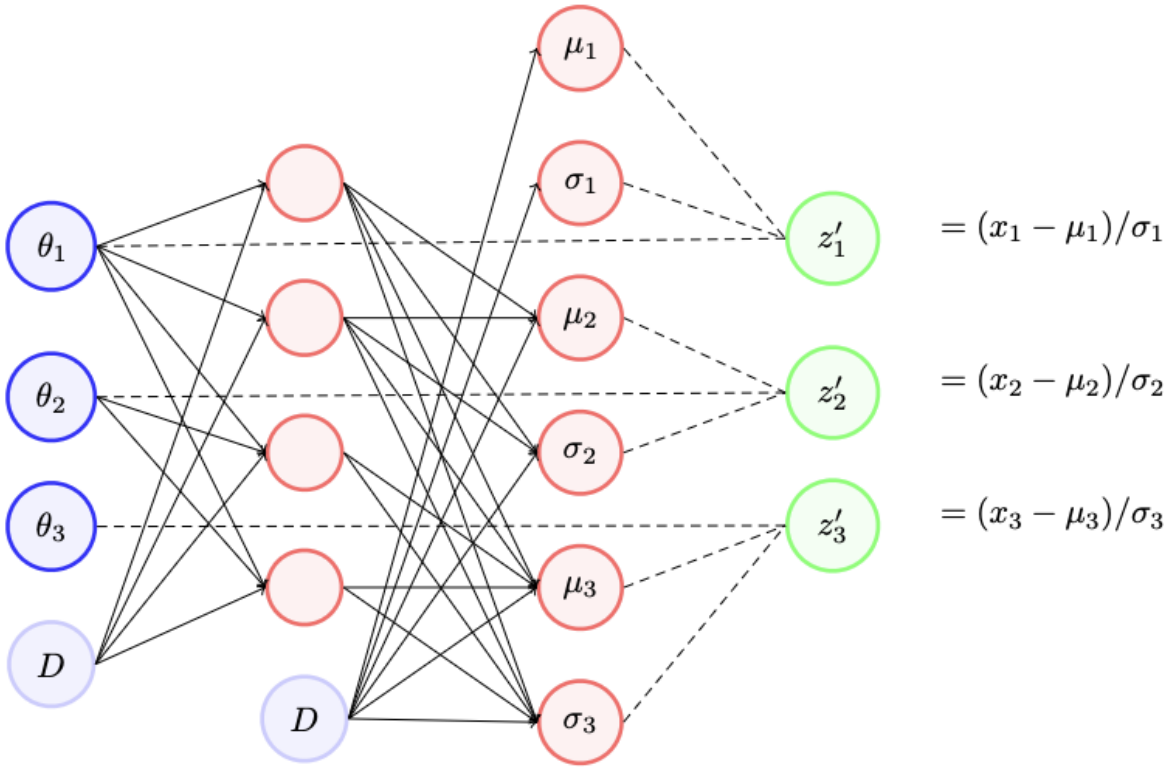
Conditional MADE



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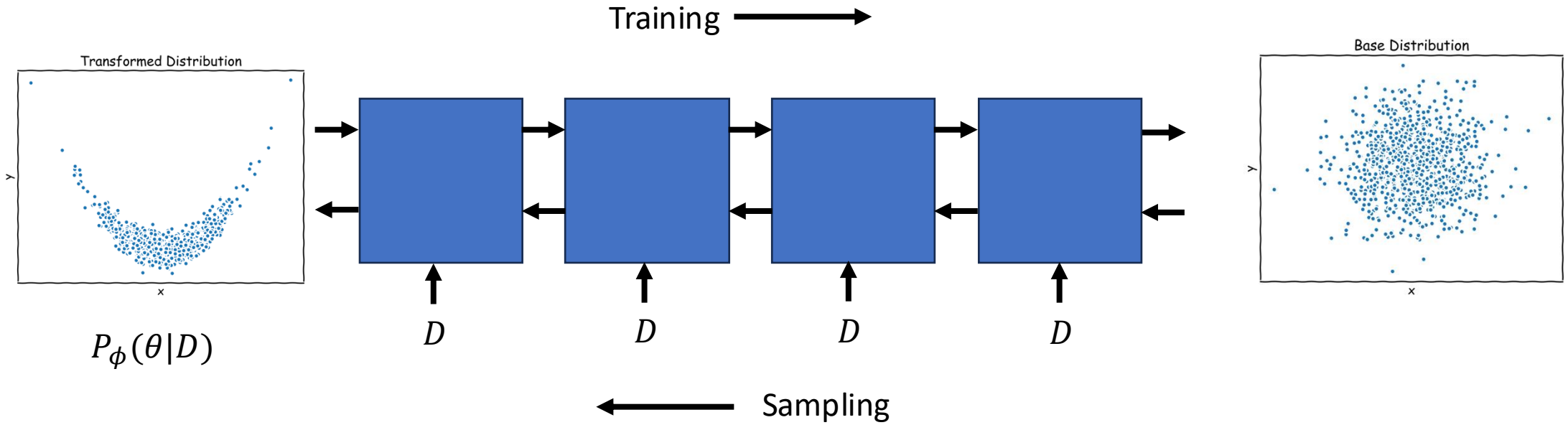
Conditional MAF



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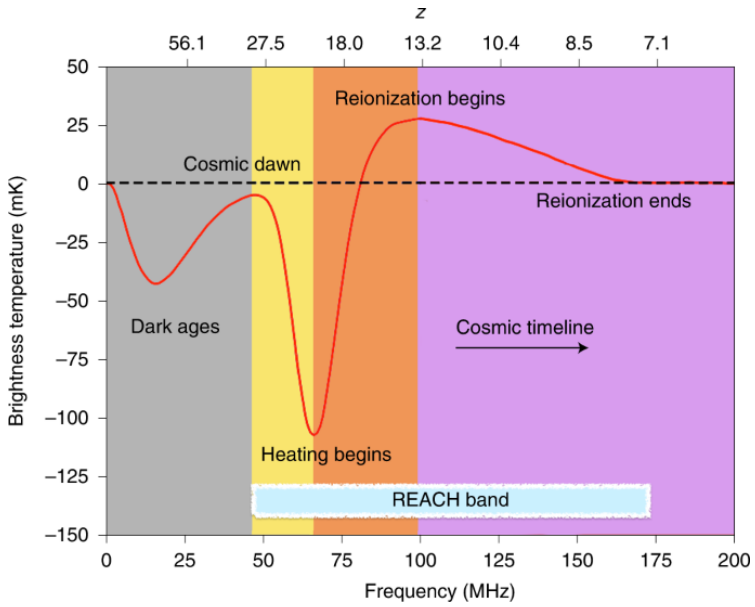
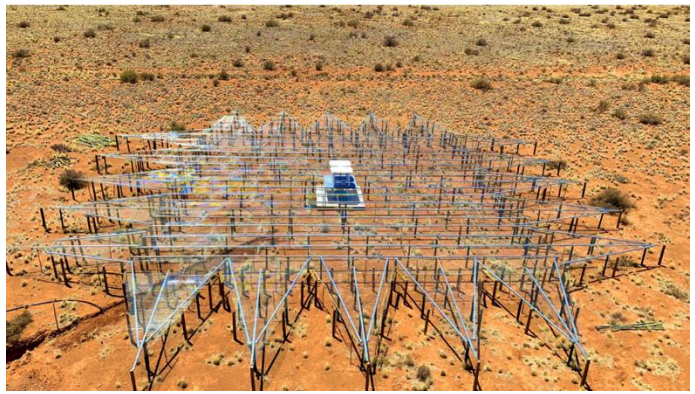
An example: 21cm Cosmology

- This time we are going to look at a toy model for the sky-averaged 21-cm signal
- Recall that this is the observable targeted by single antennas like REACH
- Signal resembles a Gaussian absorption feature so for the purposes of illustrating SBI we will model the signal as

$$T_{21}(z,\theta) = -A\exp\left(\frac{(z-z_c)^2}{\sigma^2}\right)$$





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An example: 21cm Cosmology

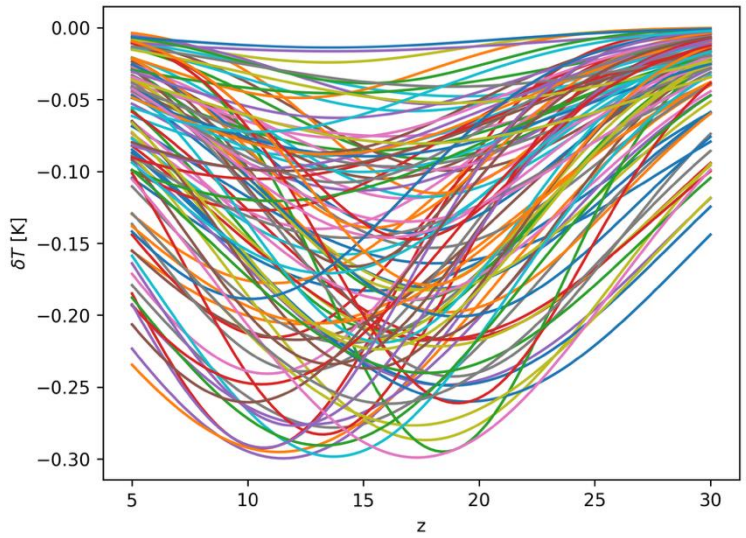
- Transforming a 3D space of

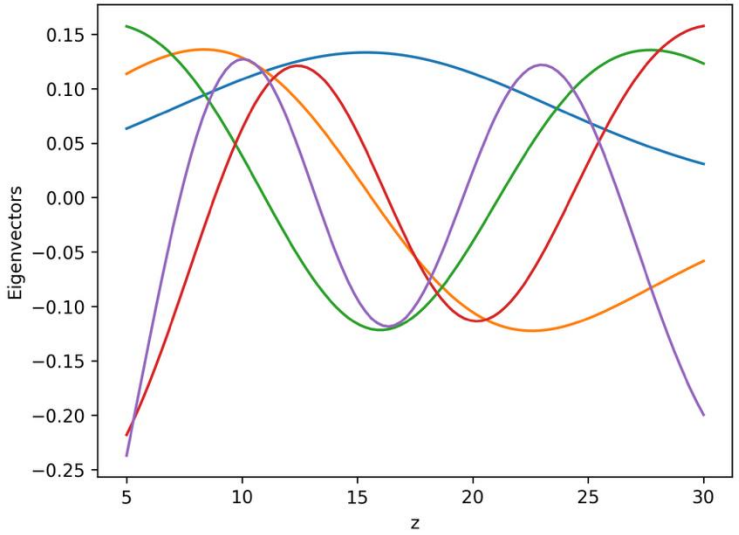
$$\theta = [A, z_c, \sigma]$$

- No noise for simplicity and 100 redshift samples

- Need to do some compression

$$f(\theta, x) = \sum_i w_i(\theta) g(x)$$





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An example: 21cm Cosmology

- Example code on https://github.com/htjb/Talks

- Quite non-trivial Tensorflow set up but I encourage you to go look at the code

- We do the usual tricks normalising our parameter and data space, optimizing the architecture of the NNs, picking the right activation functions, the right gradient descent algorithm etc…

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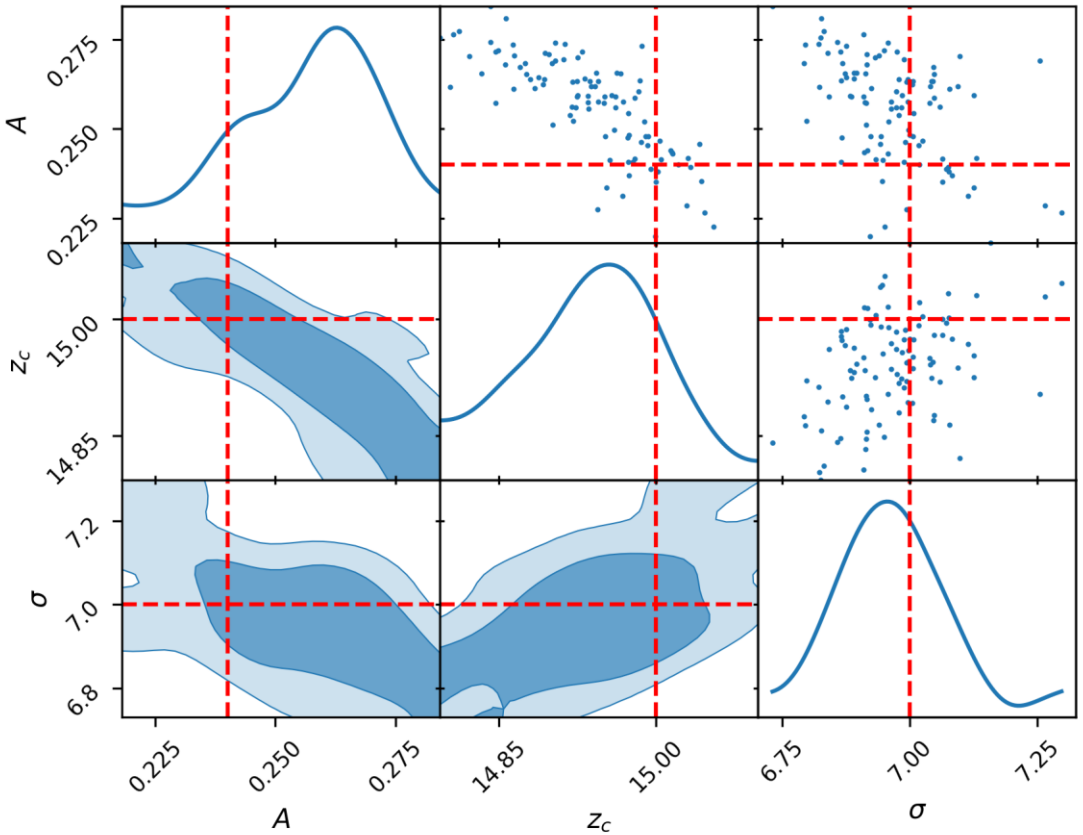
An example: 21cm Cosmology

- Once trained we pass the network samples from the base distribution and the observed data projected on to our PCA basis functions

- Returns a posterior distribution!



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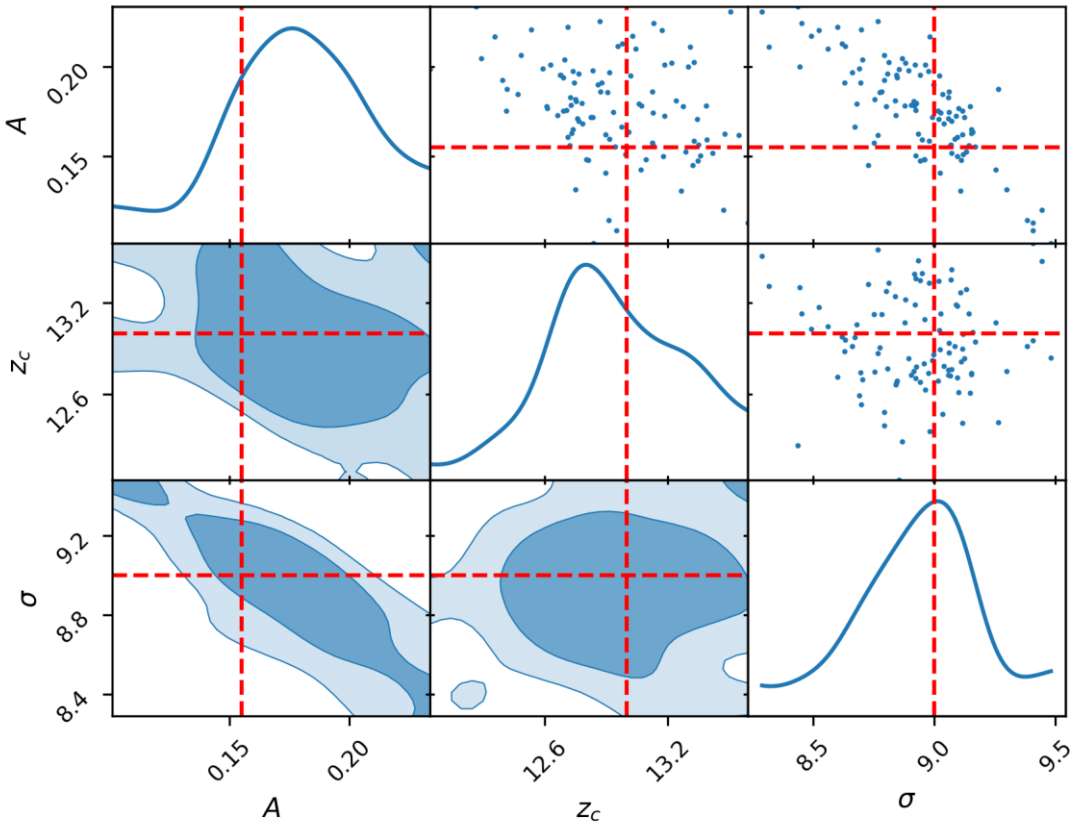
An example: 21cm Cosmology

- Once trained the NPEs are very fast
- They are amortized so we can show them another observation and recover a posterior
- But to trust the results we need to be confident our simulations capture everything in the observations!



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Neural Ratio Estimation (NRE)

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What is NRE?

- NREs are an alternative to ABC and NPE and work by predicting ratios of densities

- Typically, we set NREs up to give us the ratio $\mathrm{r} = \frac{L(\theta)}{Z}$ which we use for inference

- Note we can also use them for tension quantification (2407.15478) and model comparison (2305.11241)

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As a classification problem

- NREs are essentially classifiers
- We train with pairs of $D(\theta), \theta$ labelled with a value of 1 and $D(\theta), \phi$ labelled with a value of 0
- Use a binary cross entropy loss function

$$L = \frac{1}{N}\sum_{i} \mathrm{y_i} \log(\sigma \cdot \log r) + (1 - y_i)\log(1 - \sigma \cdot \log r)$$

where y is our label and $\log r$ is the output of the network

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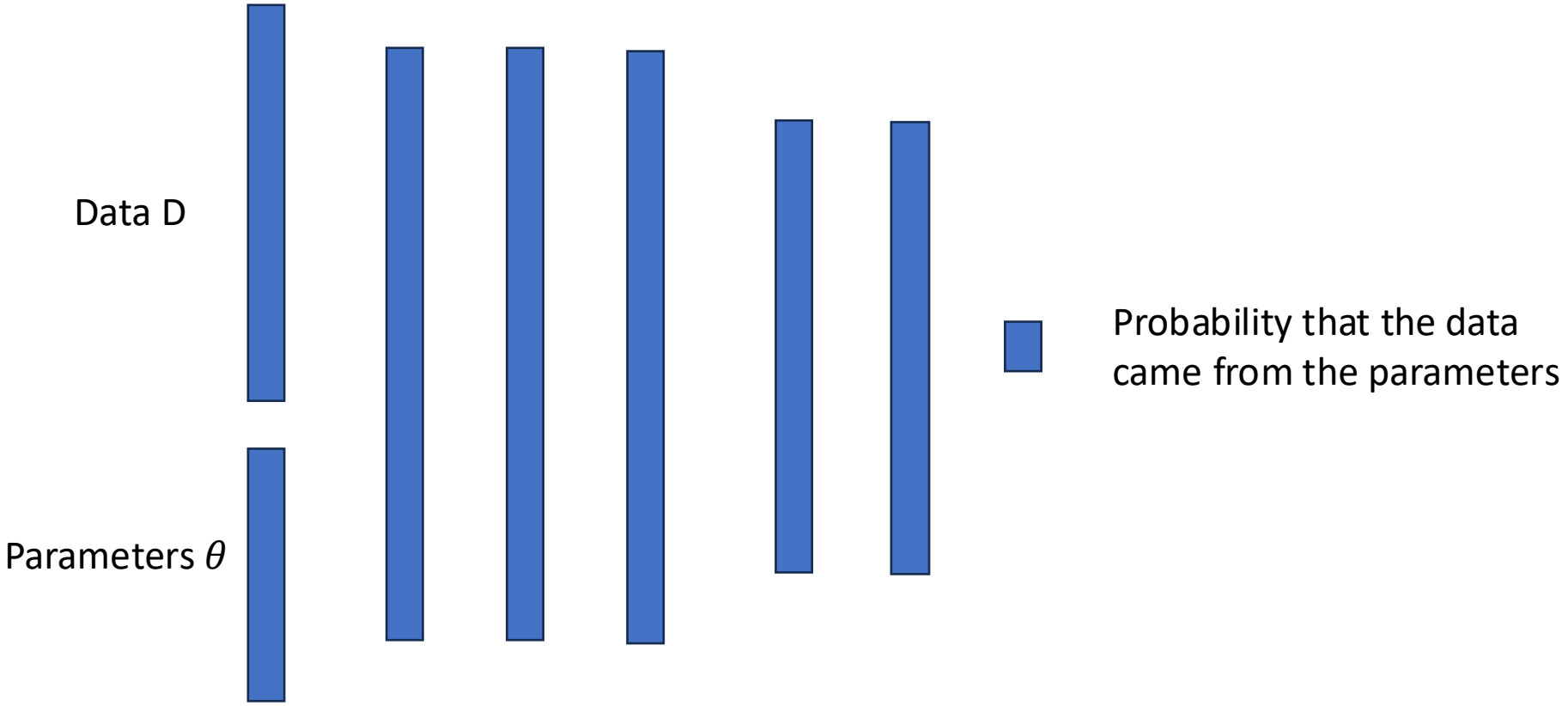
As a classification problem



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As a classification problem

- The learned output of the network is

$$r = \frac{P(D,\theta)}{P(D)P(\theta)}$$

- We can express this in more familiar language as

$$r = \frac{P(D,\theta)}{P(D)P(\theta)} = \frac{P(D|\theta)P(\theta)}{P(D)P(\theta)} = \frac{L(\theta)\pi(\theta)}{Z\,\pi(\theta)} = \frac{L(\theta)}{Z}$$

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As a classification problem

- The network learns what kind of data different sets of parameters produce

- Once trained we feed the network a real data set and samples of θ from a prior to recover a posterior as

$$P(\theta|D) = r\,\pi(\theta) = \frac{L(\theta)}{Z}\,\pi(\theta)$$

- NREs are amortized like NPEs and so can be reused on many different observations

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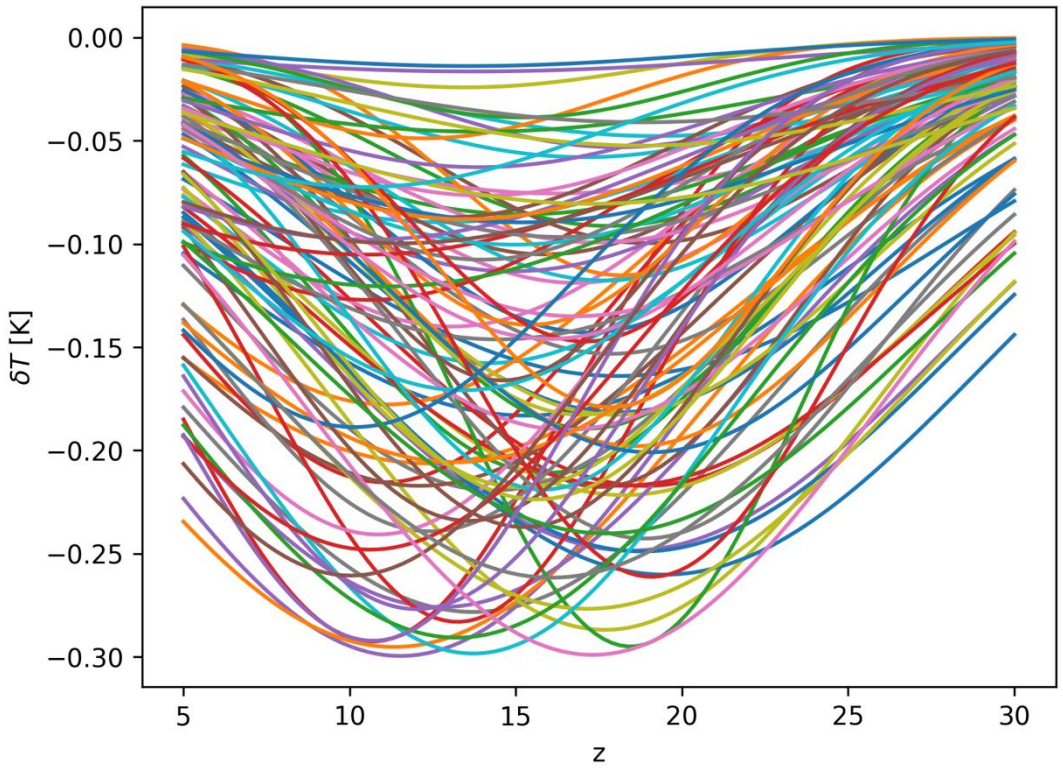
An Example: 21cm Cosmology




```
norm_signals = (signals - signals.mean())/signals.std()
norm_params = (theta- theta.mean(axis=0))/theta.std(axis=0)

norm_data = np.hstack([norm_signals, norm_params])
norm_labels = np.ones(n)

idx = np.arange(0, len(norm_data))
random.shuffle(idx)
shuffled_params = norm_params[idx, :]
shuffled_data = np.hstack([norm_signals, shuffled_params])
shuffled_labels = np.zeros(n)

data = np.vstack([shuffled_data, norm_data])
labels = np.hstack([shuffled_labels, norm_labels])

idx = np.arange(0, 2*n)
random.shuffle(idx)
data = data[idx, :]
labels = labels[idx]
```

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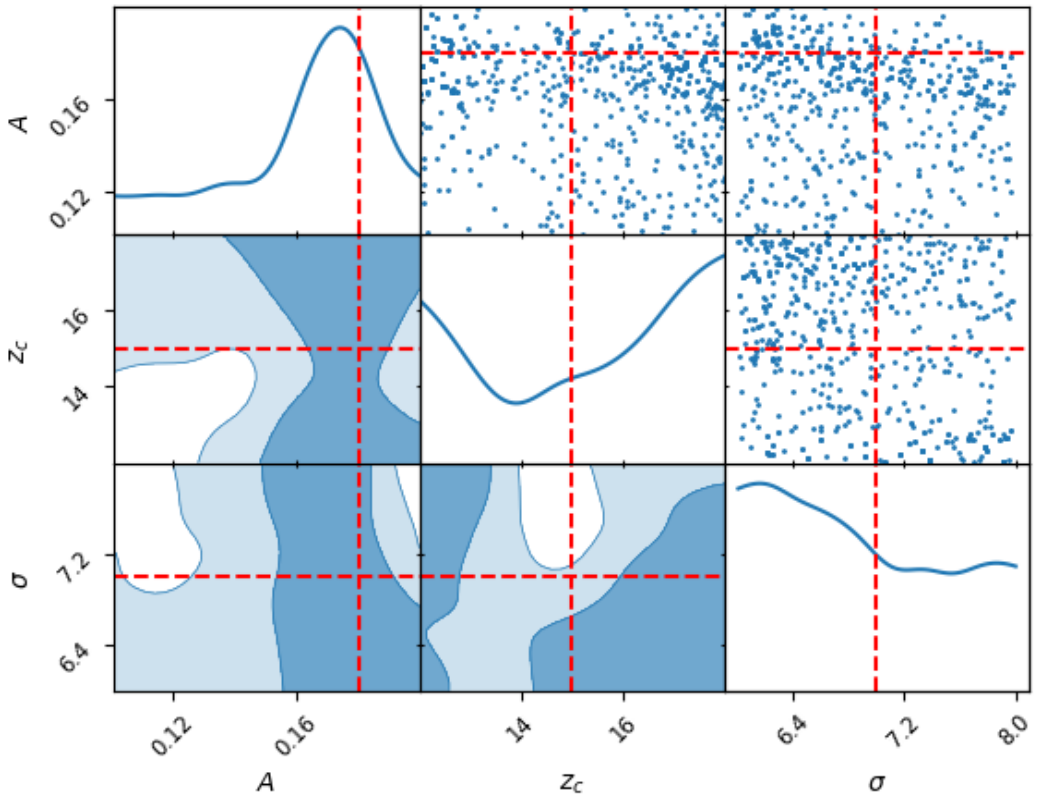
An Example: 21cm Cosmology

- Train the NRE
- Sample a prior and evaluate $P(\theta|D)$ using the NRE
- Here I am just sampling uniformly across the prior
- But really want to wrap this inside an MCMC algorithm like Nested Sampling or emcee



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Summary

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Simulation Based Inference

- Advantageous when we cannot analytically write a likelihood function or we need access to posteriors very quickly
- Strongly dependent on the accuracy of our simulations
- We are still learning about how SBI methods scale to higher dimensions and cope with model misspecification

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Join the research group!

- You've heard a lot from a lot of people during this course!
- We are always interested in prospective PhD students
- The group works on a diverse range of topics so hopefully we are doing something that interests you!
- Related opportunities in Will Handley's Group and Anastasia Fialkov's Group
- Remember there are code examples on my github

https://www.cavendishradiocosmology.com/
https://handley-lab.co.uk/
https://www.cosmicdawnlab.com/